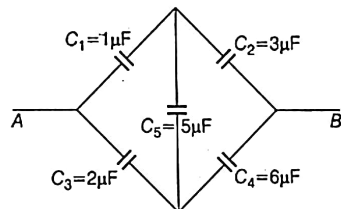


The given circuit diagram can be represented as

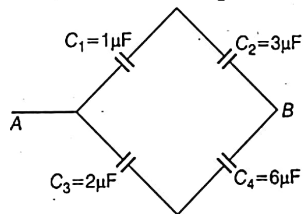


Since, $\frac{C_1}{C_2} = \frac{C_3}{C_4} = \frac{1}{3}$

Hence, given circuit follows the condition of balance wheatstone bridge.

Hence, capacitor C_5 is ineffective.

Now, the circuit diagram is redrawn as



$$C_{AB} = \frac{C_1 C_2}{C_1 + C_2} + \frac{C_3 C_4}{C_3 + C_4} = \frac{1 \times 3}{1 + 3} + \frac{2 \times 6}{2 + 6}$$

$$= \frac{3}{4} + \frac{12}{8} = \frac{9}{4} \mu\text{F}$$



4%

2.10% marked this



6%

Correct

You Marked

97.59% marked this

EVALUATION



TIME TAKEN

You: 50 sec

Others: 43 sec

TIME ALLOTTED

90 sec

DIFFICULTY

Easy

CHAPTER

Mathematics in Physics

TOPIC

Errors of Measurement

DIDN'T ATTEMPT

4.62%

GOT IT RIGHT

97.59%

GOT IT WRONG

2.41%

SOLUTION

Video Solution

$$Q = I^2 R t$$

Required error is $2 \times 2\% + 1\% + 1\% = 6\%$

Prev

2



Next

Activate Windows

Go to Settings to activate Windows.

DIFFICULTY

Medium

CHAPTER

Electromagnetic Waves

TOPIC

Electromagnetic Waves and Maxwell's equations

DIDN'T ATTEMPT 

9.23%

GOT IT RIGHT 

97.37%

GOT IT WRONG 

2.63%

SOLUTION

 Video Solution

Relation between electric and magnetic field in electromagnetic wave

$$B = \frac{E}{c}$$

$$E = 750 \text{ N/C}$$

Speed of light,

$$c = 3 \times 10^8 \text{ m/s}$$

$$B = \frac{750}{3 \times 10^8}$$

$$B = 2.5 \times 10^{-6} \text{ T}$$

As, B must be perpendicular to both $\mathbf{c}$ and $\mathbf{E}$, i.e. it is along Z -axis.

$$B = 2.5 \times 10^{-6} \hat{k} \text{ T}$$

 Prev

3

v

Next Activate Windows
Go to Settings to activate Windows.

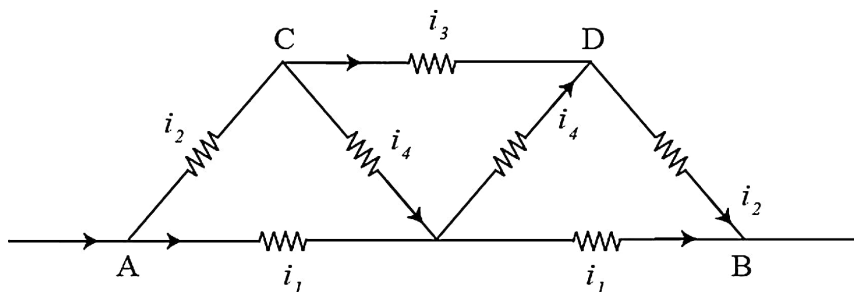
At A current is distributed and at B currents are collected. Between A and B, the distribution is symmetrical. It has been shown in the figure. It appears that current in AO and OB remains same. At O, current i_4 returns back without any change. If we detach O from AB there will not be any change in distribution. Now, CO & OD will be in series hence its total resistance = 2Ω

It is in parallel with CD, so, equivalent resistance

$$= \frac{2 \times 1}{2+1} = \frac{2}{3} \Omega$$

This equivalent resistance is in series with AC & DB so, total resistance

$$= \frac{2 \times 1}{2+1} = \frac{2}{3} \Omega$$



Now $\frac{8}{3} \Omega$ is parallel to AB, that is, 2Ω , so total resistance = $\frac{8/3 \times 2}{8/3 + 2} = \frac{16/3}{14/3} = \frac{16}{14} = \frac{8}{7} \Omega$

SOLUTION

[▶ Video Solution](#)

The radius of the hydrogen atom,

$$r_n = r_0 n^2$$

$$\Rightarrow r_{n+1} = r_0(n+1)^2, r_n = r_0(n)^2 \text{ \& } r_{n-1} = r_0(n-1)^2$$

$$\Rightarrow r_{n+1} - r_n = r_{n-1}$$

$$\Rightarrow r_0(n+1)^2 - r_0(n)^2 = r_0(n-1)^2$$

$$\Rightarrow (n+1)^2 - (n)^2 = (n-1)^2$$

$$\Rightarrow 2n+1 = n^2 + 1 - 2n$$

$$\Rightarrow 4n = n^2$$

$$\Rightarrow n = 4$$

now Recall the Bohr's second postulate,

$$mvr = \vec{J} = \frac{nh}{2\pi} = \frac{4h}{2\pi} = \frac{2h}{\pi}, \text{ where } \vec{J} \text{ is known as total angular momentum,}$$

[◀ Prev](#)

5

✓

[Next ▶](#)

Why did you get the Question Wrong?

Please select the reason why you attempted the question wrong, this will help us in providing more accurate analysis.



Silly Mistake



Calculation Error



Time Pressure



Conceptual Gap



Misinterpreted Question



Forgot Formula



Overthinking



Missed a Step

EVALUATION

Incorrect

TIME TAKEN

You: 105 sec

Others: 50 sec

TIME ALLOTTED

90 sec

DIFFICULTY

Easy

CHAPTER

Kinetic Theory of Gases

TOPIC

Speed of Gas

DIDN'T ATTEMPT ☐

17.45%

GOT IT RIGHT ☒

52.76%

GOT IT WRONG ☒

47.24%

SOLUTION

[▶ Video Solution](#)

According to Maxwell's velocity distribution curve, maximum molecules have same velocity which is called most probable velocity.

[< Prev](#)

6



[Next >](#)

SOLUTION

[▶ Video Solution](#)

We know that the centripetal force acting on a body in any orbit is provided by gravitational force.

Now according to the question gravitational force between two masses m_1 and m_2 is, $F_G = \frac{Gm_1m_2}{R}$, where G is the gravitational constant, R is separation between them.

Now, let m_1 and m_2 be the mass of two bodies, and m_2 is orbiting around m_1 and $m_2 \gg m_1$.

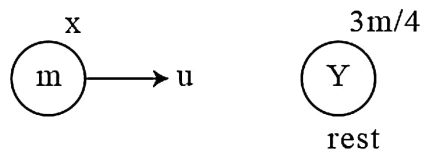
$$\therefore F_G = \frac{Gm_1m_2}{R} = \frac{m_2V^2}{R}$$

where V is the orbital velocity.

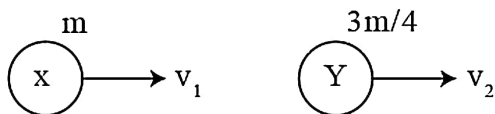
$$\Rightarrow V^2 = Gm_1$$

$\therefore V$ is proportional to R^0 .

SOLUTION

[▶ Video Solution](#)

JBC



JAC

$$* \quad mu + \frac{3m}{4}(0) = mv_1 + \frac{3m}{4}v_2$$

$$* \quad u - 0 = 1(v_2 - v_1)$$

$$\therefore v_1 = \frac{u}{7}, v_2 = \frac{8u}{7}$$

$$\frac{\lambda_y}{\lambda_x} = \frac{h / \frac{3m}{4} \cdot \frac{8u}{7}}{h / m \cdot \frac{u}{7}} = \frac{1}{6}$$

[◀ Prev](#)

8

v

[Next ▶](#)

EVALUATION



TIME TAKEN

You: 198 sec

Others: 114 sec

TIME ALLOTTED

90 sec

DIFFICULTY

Easy

CHAPTER

Motion In One Dimension

TOPIC

Graphs of motion in one dimension

DIDN'T ATTEMPT ☐

19.12%

GOT IT RIGHT ☒

76.57%

GOT IT WRONG ☐

23.43%

SOLUTION

[▶ Video Solution](#)

We can define the average speed of the particle as follows:

$v_{\text{avg}} = \int_{t_1}^{t_2} a \, dt$ (we have not used the modulus sign here since both the functions are positive in their domains).

The above equation suggests us that the average speed equals the area under the acceleration-time curve. As per the given graph, the areas under both the curves are different and hence, the average speed is also different.

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9

[Next >](#)

Given that, in a closed system,
initially $Q_1 = 200$ kJ of heat supplied at constant volume, i.e., work done in this case is,

$$\Delta W_1 = P(\Delta V) = 0.$$

From the first law of thermodynamics,

$$Q_1 = \Delta U_1 + \Delta W_1 \Rightarrow \Delta U_1 = Q_1 = 200 \text{ kJ}$$

Next, the system ejects 100 kJ of heat by doing 50 kJ of work on the system.

$$\text{i.e., } Q_2 = -100 \text{ kJ and } \Delta W_2 = -50 \text{ kJ}$$

From the first law of thermodynamics,

$$Q_2 = \Delta U_2 + \Delta W_2 \Rightarrow -100 = \Delta U_2 - 50$$

$$\Delta U_2 = -100 + 50 \Rightarrow \Delta U_2 = -50 \text{ kJ}$$

Finally, to restore the system to initial state an adiabatic process is found.

In the adiabatic process, $\Delta Q = 0$.

From the first law of thermodynamics,

$$Q_3 = \Delta U_3 + \Delta W_3 \Rightarrow \Delta U_3 + \Delta W_3 = 0$$

$$\Rightarrow \Delta U_3 = -\Delta W_3$$

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v

[Next >](#)

In a closed system, the total change in internal energy of the system is zero, i.e.,

$$Q_2 = \Delta U_2 + \Delta W_2 \Rightarrow -100 = \Delta U_2 - 50$$

$$\Delta U_2 = -100 + 50 \Rightarrow \Delta U_2 = -50 \text{ kJ}$$

Finally, to restore the system to initial state an adiabatic process is found.

In the adiabatic process, $\Delta Q = 0$.

From the first law of thermodynamics,

$$Q_3 = \Delta U_3 + \Delta W_3 \Rightarrow \Delta U_3 + \Delta W_3 = 0$$

$$\Rightarrow \Delta U_3 = -\Delta W_3$$

In a closed system, the total change in internal energy of the system is zero, i.e.,

$$\Delta U_1 + \Delta U_2 + \Delta U_3 = 0$$

$$200 \text{ kJ} + (-50 \text{ kJ}) + (-\Delta W_3) = 0$$

$$\Rightarrow \Delta W_3 = 150 \text{ kJ}$$

Hence, work done by the system during the adiabatic process is 150 kJ.

DIDN'T ATTEMPT ☐
47.54%GOT IT RIGHT ☒
90.47%GOT IT WRONG ☐
9.53%

SOLUTION

[▶ Video Solution](#)

Here, $a = 1 \text{ cm} = 0.01 \text{ m}$. The mass will remain in contact with surface,

$$\text{if } mg = m\omega^2 a$$

$$\text{or } \omega = \sqrt{\frac{g}{a}}$$

$$\text{or } 2\pi v = \sqrt{\frac{g}{a}}$$

Or

$$v = \frac{1}{2\pi} \sqrt{\frac{g}{a}}$$

$$= \frac{7}{2 \times 22} \sqrt{\frac{980}{1}}$$

$$= 4.9 \text{ Hz} \approx 5 \text{ Hz}$$

$$= 4.9 \text{ Hz} \approx 5 \text{ Hz}$$

~

[< Prev](#)

11

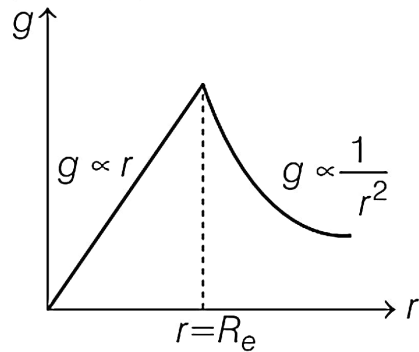
v

[Next >](#)

At height h ,

$$g' = g \left(\frac{R_e^2}{(R_e + h)^2} \right) = \frac{gR_e^2}{r^2} \Rightarrow g' \propto \frac{1}{r^2}$$

So, correct graph is

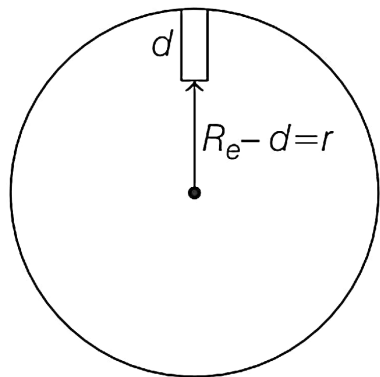


*

At depth d ,

$$g' = g \left(1 - \frac{d}{R_e} \right) = g \left(\frac{R_e - d}{R_e} \right) = \frac{gr}{R_e}$$

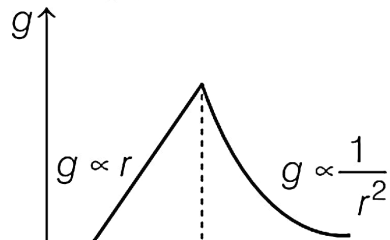
$\Rightarrow g' \propto r$ (where, r = distance from centre of earth)



At height h ,

$$g' = g \left(\frac{R_e^2}{(R_e + h)^2} \right) = \frac{gR_e^2}{r^2} \Rightarrow g' \propto \frac{1}{r^2}$$

So, correct graph is



← Prev

12

✓

Next →

EVALUATION



TIME TAKEN

You: 284 sec

Others: 201 sec

TIME ALLOTTED

120 sec

DIFFICULTY

Medium

CHAPTER

Electromagnetic Induction

TOPIC

Self and Mutual Inductance

DIDN'T ATTEMPT ☐

36.33%

GOT IT RIGHT ☒

70.99%

GOT IT WRONG ☐

29.01%

SOLUTION

[▶ Video Solution](#)

Given, Area of the inner solenoid, $A = 2 \times 10^{-3}$

Mutual inductance of the system, $M = \mu_0 n_1 n_2 A l$

$$\Rightarrow M = 4\pi \times 10^{-7} \times 1000 \times 4000 \times 2 \times 10^{-3} \times 60 \times 10^{-3}$$

$$\Rightarrow M = 9 \text{ mH}$$

^

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13

v

[Next >](#)

Please select the reason why you attempted the question

wrong, this will help us in providing more accurate analysis.

- ☐ Silly Mistake ☐ Calculation Error ☐ Time Pressure ☐ Conceptual Gap
- ☐ Misinterpreted Question ☐ Forgot Formula ☐ Overthinking ☐ Missed a Step

EVALUATION

Incorrect

TIME TAKEN

You: 84 sec
Others: 53 sec

TIME ALLOTTED

120 sec

DIFFICULTY

Medium

CHAPTER

Alternating Current

TOPIC

Alternating Current, Voltage and Power

DIDN'T ATTEMPT

17.87%

GOT IT RIGHT

80.05%

GOT IT WRONG

19.95%

SOLUTION

[▶ Video Solution](#)

For ideal inductors and capacitors there will be no dissipation of energy. They first take energies and then release them again for an AC circuit. But in resistors there is dissipation of energy which equals $i^2 R$. Average power consumed in an ac circuit is equal to average power consumed by resistors in the circuit.

~

[< Prev](#)

14

▼

[Next >](#)

Activate Windows
Go to Settings to activate Windows.

Please select the reason why you attempted the question wrong, this will help us in providing more accurate analysis.

- ☐ Silly Mistake ☐ Calculation Error ☐ Time Pressure ☐ Conceptual Gap ☒ Misinterpreted Question ☐ Forgot Formula ☐ Overthinking ☐ Missed a Step

EVALUATION

Incorrect

TIME TAKEN

You: 84 sec
Others: 53 sec

TIME ALLOTTED

120 sec

DIFFICULTY

Medium

CHAPTER

Alternating Current

TOPIC

Alternating Current, Voltage and Power

DIDN'T ATTEMPT

17.87%

GOT IT RIGHT

80.05%

GOT IT WRONG

19.95%

SOLUTION

[▶ Video Solution](#)

For ideal inductors and capacitors there will be no dissipation of energy. They first take energies and then release them again for an AC circuit. But in resistors there is dissipation of energy which equals $i^2 R$. Average power consumed in an ac circuit is equal to average power consumed by resistors in the circuit.

~

[< Prev](#)

14

▼

[Next >](#)

Activate Windows
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For projectile motion, horizontal range is given by,

$$R = \frac{u^2 \sin 2\theta}{g}$$

Substituting $\theta = 45^\circ$ gives the maximum range.

$$R_R = \frac{u^2 \times \sin(2 \times 45^\circ)}{g} = \frac{u^2}{g}$$

The ranges of projectile for different angles of projection are calculated as follows,

$$R_P = \frac{u^2 \times \sin(2 \times 15^\circ)}{g} = \frac{u^2}{2g} = 0.5 \frac{u^2}{g},$$

$$R_Q = \frac{u^2 \times \sin(2 \times 30^\circ)}{g} = \frac{\sqrt{3}u^2}{2g} = 0.866 \frac{u^2}{g}$$

and

$$R_S = \frac{u^2 \times \sin(2 \times 60^\circ)}{g} = \frac{\sqrt{3}u^2}{2g}$$

Hence, range increases when θ increases from 0° to 45° but, it decreases beyond 45° . When the angle of projection is very far from 45° , then, range will be minimum which is 15° for body P .



Statement -1 is true, Statement-2 is false

Correct

You Marked

75.57% marked this



Statement -1 is false, Statement-2 is true

1.57% marked this

EVALUATION

Correct

TIME TAKEN

You: 57 sec

Others: 46 sec

TIME ALLOTTED

120 sec

DIFFICULTY

Hard

CHAPTER

Magnetic Effects of Current

TOPIC

Force and Torque on Current Carrying Conductor

DIDN'T ATTEMPT ☐

26.44%

GOT IT RIGHT ☒

75.57%

GOT IT WRONG ☐

24.43%

SOLUTION

[▶ Video Solution](#)

Placing a soft iron core inside the coil makes the magnetic field radial which increase the torque acting on the coil due a give current flowing in the coil. The sensitivity increase because, for the same current, the deflection of the coil is more when a core is inserted inside the coil. Statement-1 is true and Statemenet-2 is false.

;

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[Next >](#)

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You: 93 sec
Others: 50 sec

120 sec

DIFFICULTY
Medium

CHAPTER
Semiconductors

TOPIC
Diodes

DIDN'T ATTEMPT 
7.01%

GOT IT RIGHT 
96.04%

GOT IT WRONG 
3.96%

SOLUTION

 Video Solution

In circuit A , both $(p - n)$ junction diode act as forward biasing. Hence, current flows in circuit A .

Total resistance R is given by

$$\frac{1}{R} = \frac{1}{4} + \frac{1}{4}$$

$$\text{or } \frac{1}{R} = \frac{2}{4}$$

$$\text{or } R = 2\Omega$$

According to Ohm's law

$$V = I_A R$$

$$\text{or } 8 = I_A \times 2$$

$$\text{or } I_A = 4 \text{ A}$$

;

 Prev

17



Next 

Activate Windows
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Kinetic energy

$$K = \frac{1}{2} m u_1^2$$

$$u_1 = \sqrt{\frac{2K}{m}}$$

$$\frac{1}{2} m v_1^2 = \frac{K}{9}$$

$$v_1^2 = \frac{2K}{9m}$$

$$\text{Velocity, } v_1 = \sqrt{\frac{2K}{9m}}$$

$$\text{Kinetic energy } K = \frac{p^2}{2m}$$

From conservation of the momentum

$$m u_1 = -m v_1 + p_B$$

$$m(u_1 + v_1) = p_B$$

Final momentum of B

$$p_B = m \left[\sqrt{\frac{2K}{m}} + \sqrt{\frac{2K}{9m}} \right]$$

$$= \sqrt{2mK} + \sqrt{\frac{2mK}{3}} = p + \frac{p}{3} = \frac{4p}{3}$$

;

◀ Prev

18

▼

Next ▶

EVALUATION

Incorrect

TIME TAKEN

You: 128 sec

Others: 98 sec

TIME ALLOTTED

90 sec

DIFFICULTY

Easy

CHAPTER

Mechanical Properties of Fluids

TOPIC

Surface tension

DIDN'T ATTEMPT ☐

19.72%

GOT IT RIGHT ☒

88.71%

GOT IT WRONG ☐

11.29%

SOLUTION

[▶ Video Solution](#)

We have $\Delta U = (\text{surface energy of bigger drop}) - (\text{surface energy of lower drop})$

$$\Delta U = (T \times 4\pi R^2) - (nT \times 4\pi r^2)$$

$$|\Delta U| = (4\pi r^2 n - 4\pi R^2)T$$

~

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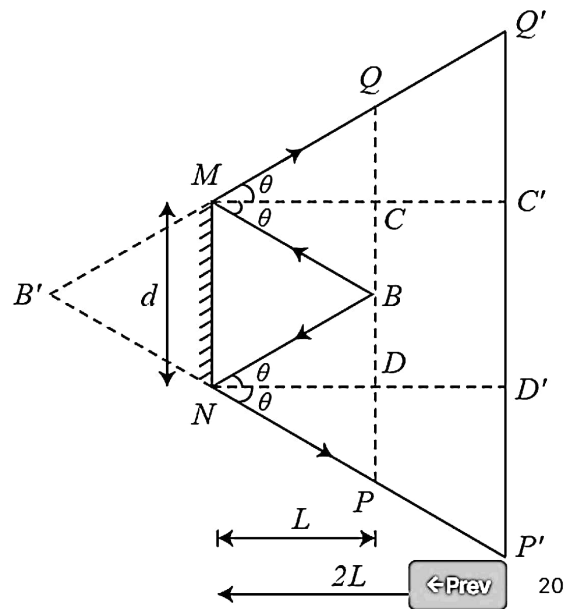
✓

[Next >](#)

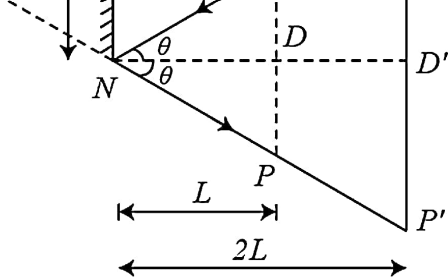
Refer to the diagram shown below, the image of the light source placed at the centre of the mirror B is formed at B'.

The extreme reflected rays meet the vertical line at distance $2L$ from the mirror at Q' and P'.

It is clear that the man can observe in the angular region P'BQ'. The observer can see through the vertical distance P'Q' at distance $2L$ from the mirror, where reflected ray from B' can meet the line.



Using the geometry, $CD = d = C'D'$



Using the geometry, $CD = d = C'D'$

Also, $PD = BD = \frac{d}{2}$ and $BC = CQ = \frac{d}{2}$

Hence, distance through which the man can observe the image
 $= Q'C' + C'D' + D'P'$

$$= 2 \times QC + C'D' + 2 \times DP$$

$$= 2 \times \frac{d}{2} + d + 2 \times \frac{d}{2} = 3d$$

!

DIDN'T ATTEMPT 

25.00%

GOT IT RIGHT 

90.70%

GOT IT WRONG 

9.30%

SOLUTION

 Video Solution

Equation of wave, $y = 0.2 \sin(1.5x + 60t)$ Comparing with standard equation,

$$y = A \sin(kx + \omega t)$$

$$k = 1.5, \omega = 60 \text{ rad/s}$$

$$\therefore \text{Velocity of wave, } v = \frac{\omega}{k} = \frac{60}{1.5} = 40 \text{ m/s}$$

Velocity of wave in a stretched string,

$$v = \sqrt{\frac{T}{m}}$$

where m = linear density

T = tension in the string

So,

$$T = v^2 m$$

$$= (40)^2 \times 3 \times 10^{-4} = 0.48$$

 Prev

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v

Next 

Activate Windows

Go to Settings to activate Windows.

Given, mass of disc = m

Radius of disc = a

Angular speed = ω

Moment of inertia of the disc, $I = \frac{1}{2} ma^2$

Initial angular momentum, $L_i = I \times \omega = \frac{1}{2} ma^2 \omega$

Let ω' be the angular momentum of the disc when the particle of mass m is attached at the rim.

Final angular momentum, $L_f = I\omega' + mav'$

$$= \frac{1}{2} ma^2 \omega' + ma^2 \omega'$$

$$[\because v' = \omega' a]$$

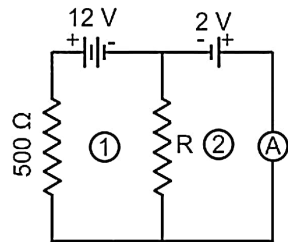
Applying conservation of angular momentum,

$$L_i = L_f$$

$$\frac{1}{2} ma^2 \omega = \frac{1}{2} ma^2 \omega' + ma^2 \omega'$$

$$\Rightarrow \omega' = \frac{\omega}{3}$$

Given circuit



In loop (1)

$$12 - 500i_1 - Ri_1 = 0$$

$$\Rightarrow 12 = i_1(500 + R) \dots(i)$$

In loop (2)

$$12 - 500i_1 - 2 = 0$$

$$\Rightarrow 10 = 500i_1$$

$$\text{or } i = \frac{1}{50} \text{ A} \dots(ii)$$

From eqs. (i) and (ii),

$$12 \times \frac{1}{i_1} = (500 + R)$$

$$\Rightarrow 12 \times 50 = 500 + R$$

$$\Rightarrow R = 100\Omega$$



You: 145 sec
Others: 152 sec

180 sec

DIFFICULTY
Medium

CHAPTER
Magnetic Effects of Current

TOPIC
Magnetic Field

DIDN'T ATTEMPT ☐
34.41%

GOT IT RIGHT ☒
91.78%

GOT IT WRONG ☐
8.22%

SOLUTION

[▶ Video Solution](#)

$$m = 0.6 \text{ g} = 0.6 \times 10^{-3} \text{ kg}, q = 25 \text{ nC} = 25 \times 10^{-9} \text{ C}$$

$$v = 1.2 \times 10^4 \text{ m/s}$$

The particle is moving with uniform velocity (acceleration is zero), so magnetic force will be balancing the weight of the particle.

$$mg = Bqv$$

$$0.6 \times 10^{-3} \times 10 = B \times 25 \times 10^{-9} \times 1.2 \times 10^4$$

$$B = \frac{0.6 \times 10^{-3} \times 10}{25 \times 10^{-9} \times 1.2 \times 10^4}$$

$$= \frac{6 \times 10^{-3}}{30 \times 10^{-5}} = \frac{6 \times 10^2}{30}$$

$$= 20 \text{ T}$$

[< Prev](#)

24



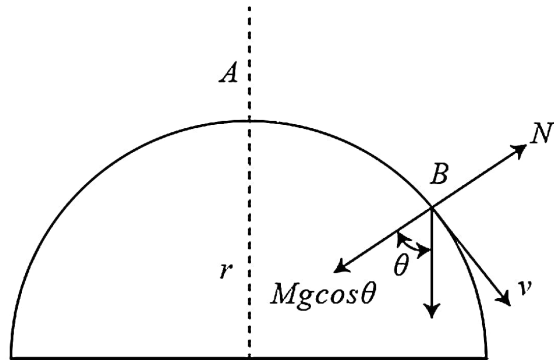
[Next >](#)

Activate Windows
Go to Settings to activate Windows.

Let the body leave the surface at point B as shown in Fig.

When the body is between points A and B , we have $Mg \cos \theta - N = \frac{Mv^2}{r}$

When the body leaves the surface at point B , the normal reaction N becomes zero.
Thus



$$Mg \cos \theta = \frac{Mv^2}{r}$$

$$\text{or } \cos \theta = \frac{v^2}{rg} = \frac{(5)^2}{5 \times 10} = \frac{1}{2} \quad \text{or} \quad \theta = 60^\circ$$

DIDN'T ATTEMPT 

31.83%

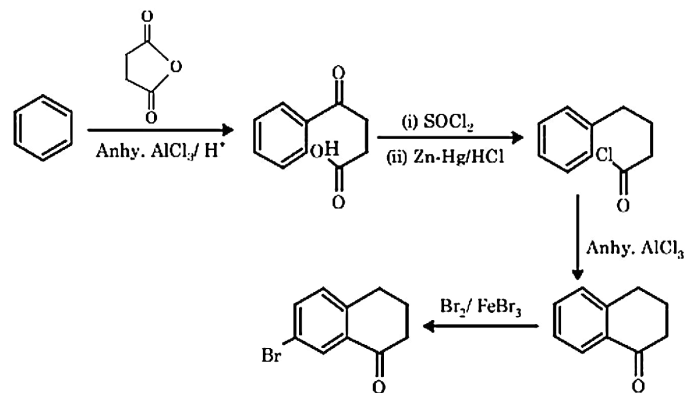
GOT IT RIGHT 

51.14%

GOT IT WRONG 

48.86%

SOLUTION

[▶ Video Solution](#)



Sodium chromate

5.23% marked this



Potassium dichromate

Correct

You Marked

89.03% marked this

EVALUATION

Correct

TIME TAKEN

You: 47 sec

Others: 44 sec

TIME ALLOTTED

120 sec

DIFFICULTY

Medium

CHAPTER

d and f Block Elements

TOPIC

Compounds of Transition Metals

DIDN'T ATTEMPT

14.27%

GOT IT RIGHT

89.03%

GOT IT WRONG

10.97%

SOLUTION

[▶ Video Solution](#)

Potassium dichromate $\text{K}_2\text{Cr}_2\text{O}_7$ is an orange-red crystalline soluble solid substance. It is a good oxidizing agent and is used in making chrome pigments, azo dyes and as a bleaching agent.

[< Prev](#)

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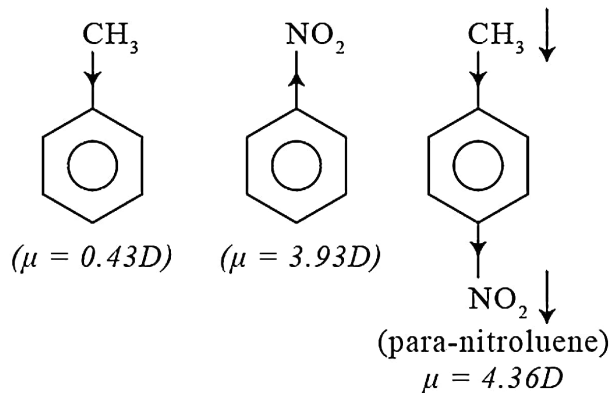
[Next >](#)

SOLUTION

[▶ Video Solution](#)

Methyl group has +I effect and $-\text{NO}_2$ group has $-I$ effect. Therefore, in p-nitro toluene the dipole moments of $-\text{CH}_3$ and $-\text{NO}_2$ groups act in the same direction. So, the resultant dipole moment is additive.

i.e., $3.93 + 0.43 = 4.36$ debye

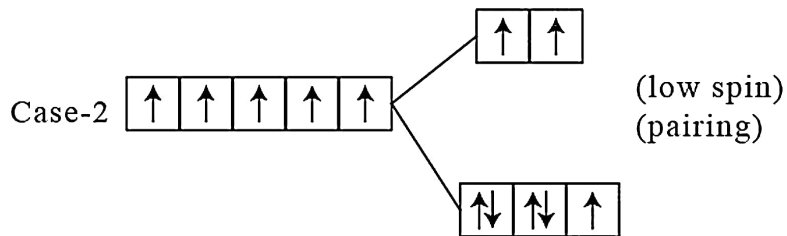
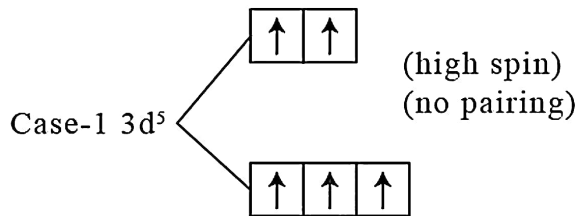


$[\text{Fe}(\text{SCN})_6]^{3-}$ and $[\text{Fe}(\text{CN})_6]^{3-}$

In both the cases the electronic configuration of Fe^{3+} will be

$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 3d^5$

Since SCN^- is a weak field ligand and CN^- is a strong field ligand, the pairing will occur only in case of $[\text{Fe}(\text{CN})_6]^{3-}$.



Case -1 $\mu = \sqrt{n(n+2)} = \sqrt{5(5+2)} = \sqrt{35} = 5.91 \text{ BM}$

Case -2 $\mu = \sqrt{n(n+2)} = \sqrt{1(1+2)} = \sqrt{3} = 1.73 \text{ BM}$

Difference in spin only magnetic moment = $5.91 - 1.73 = 4.18 \approx 4$

EVALUATION



TIME TAKEN

You: 183 sec

Others: 61 sec

TIME ALLOTTED

120 sec

DIFFICULTY

Easy

CHAPTER

Thermodynamics (C)

TOPIC

First Law and Basic Fundamentals of Thermodynamics

DIDN'T ATTEMPT ☐

18.59%

GOT IT RIGHT ☒

41.01%

GOT IT WRONG ☒

58.99%

SOLUTION

[▶ Video Solution](#) T' is final temperature.

$$q = \Delta E - W$$

$$= C_V(T' - T) + P_{\text{ext}}(2 - 1)$$

$$= C_V(T' - T) + 1 \text{ L atm}$$

$$q = \frac{3}{2} R (T' - T) + 1 \text{ L atm} = 0$$

$$T' - T = -\frac{2}{3} \frac{1}{R}$$

$$T' = T - \frac{2}{3R}$$

$$= T - \frac{2}{3 \times 0.082}$$

[< Prev](#)

30

[Next >](#)

harsh

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**FeSO₄NO₂**

7.92% marked this

**Ferrous nitrate**

9.88% marked this

EVALUATION



TIME TAKEN

You: 41 sec

Others: 40 sec

TIME ALLOTTED

90 sec

DIFFICULTY

Medium

CHAPTER

Practical Chemistry

TOPIC

Preliminary Tests

DIDN'T ATTEMPT ☐

25.18%

GOT IT RIGHT ☒

78.45%

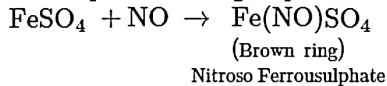
GOT IT WRONG ☐

21.55%

SOLUTION

[▶ Video Solution](#)

When NO₃⁻ is passed through FeSO₄ (neutral) and then few drops of conc. H₂SO₄ is added then the brown ring is obtained.

[< Prev](#)

31

[Next >](#)

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3

3.84% marked this



4

2.74% marked this

EVALUATION

Completed

TIME TAKEN

You: 93 sec

Others: 69 sec

TIME ALLOTTED

210 sec

DIFFICULTY

Hard

CHAPTER

Amines

TOPIC

Separation of mixture of amines

DIDN'T ATTEMPT ☐

28.96%

GOT IT RIGHT ☒

71.14%

GOT IT WRONG ☐

28.86%

SOLUTION

[▶ Video Solution](#)

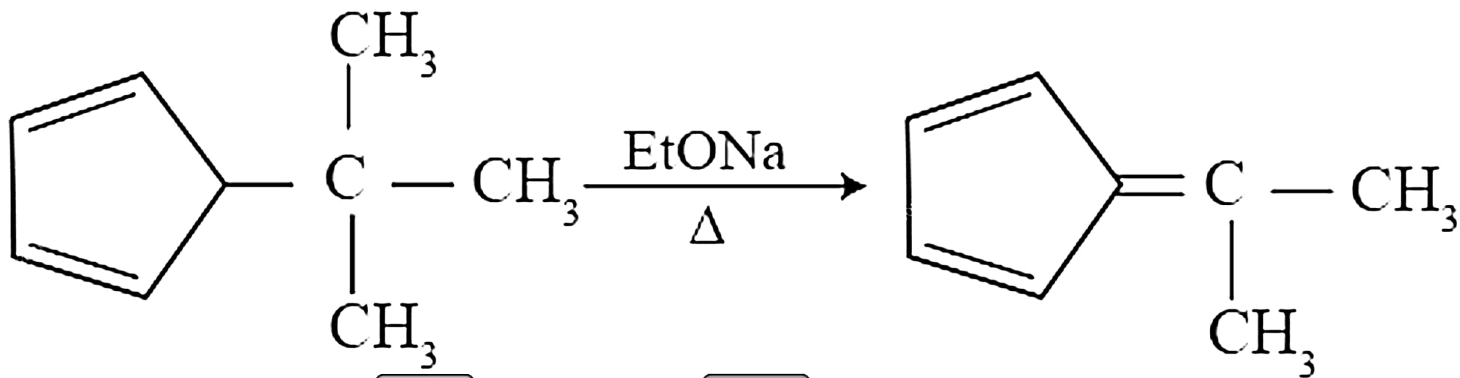
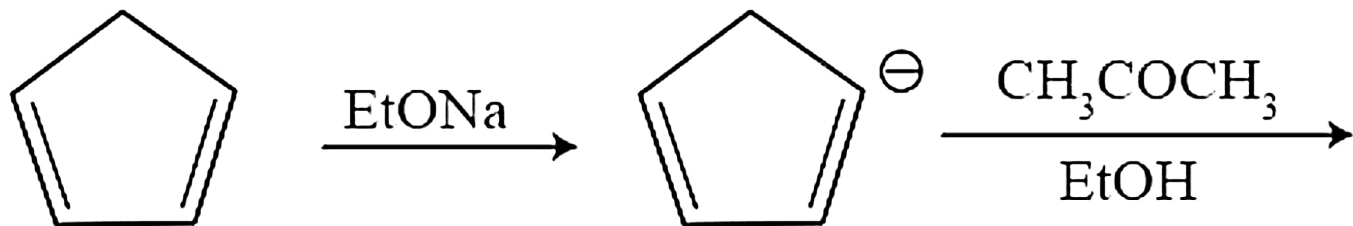
N –ethylbenzene is soluble in alkali because the hydrogen attached to nitrogen in sulphonamide is strongly acidic and forms a salt during the reactions between these two reactant molecules.

[< Prev](#)

32

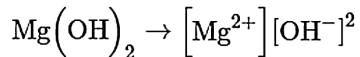


[Next >](#)



In the saturated solution of sparingly soluble electrolyte, the ionic product of ions is constant at constant temperature and it is known as Solubility Product.

Here,



Given,

$$K_{\text{sp}} \text{ for } \text{Mg}(\text{OH})_2 = 5.6 \times 10^{-12}$$

$$[\text{Mg}^{2+}] = 10^{-10} \text{ M}$$

Substituting the value, we get :

$$\begin{aligned} K_{\text{sp}} &= [\text{Mg}^{2+}][\text{OH}^-]^2 \\ \Rightarrow 5.6 \times 10^{-12} &= [10^{-10}][\text{OH}^-]^2 \\ \Rightarrow \text{OH}^- &= 0.24 \text{ M} \end{aligned}$$

Hence, the concentration of $[\text{OH}^-] = 0.24 \text{ M}$



δ -amino acids

0.45% marked this



α -amino acids

Correct

You Marked

95.76% marked this

EVALUATION



TIME TAKEN

You: 26 sec

Others: 21 sec

TIME ALLOTTED

60 sec

DIFFICULTY

Easy

CHAPTER

Biomolecules

TOPIC

Proteins

DIDN'T ATTEMPT ☐

17.93%

GOT IT RIGHT ☒

95.76%

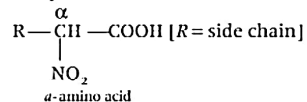
GOT IT WRONG ☐

4.24%

SOLUTION

[▶ Video Solution](#)

Only α -amino acids are obtained by hydrolysis of proteins.



[< Prev](#)

35



[Next >](#)

Activate Windows
Go to Settings to activate Windows.



High performance liquid chromatography (HPLC)

12.92% marked this



Thin layer chromatography (TLC)

32.21% marked this

EVALUATION

Noted for review

TIME TAKEN

You: 81 sec

Others: 33 sec

TIME ALLOTTED

90 sec

DIFFICULTY

Medium

CHAPTER

General Organic Chemistry

TOPIC

Purification of Organic Compounds

DIDN'T ATTEMPT ☐

34.71%

GOT IT RIGHT ☒

52.68%

GOT IT WRONG ☐

47.32%

SOLUTION

[▶ Video Solution](#)

In paper chromatography both the mobile and stationary phases are liquids. The stationary phase liquid would be an immiscible liquid with the mobile phase. The stationary phase liquid would be an immiscible liquid with the mobile phase.

Column and thin layer chromatography are illustrations of liquid/ solid chromatography where the mobile phase is a liquid while the stationary phase is a solid.

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DIDN'T ATTEMPT ☐

17.93%

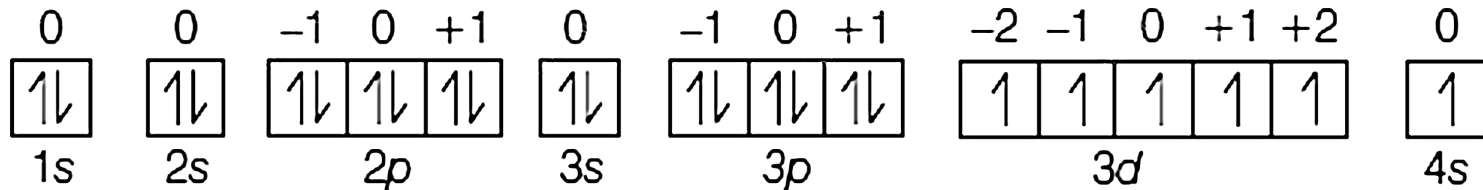
GOT IT RIGHT ☒

82.24%

GOT IT WRONG ☐

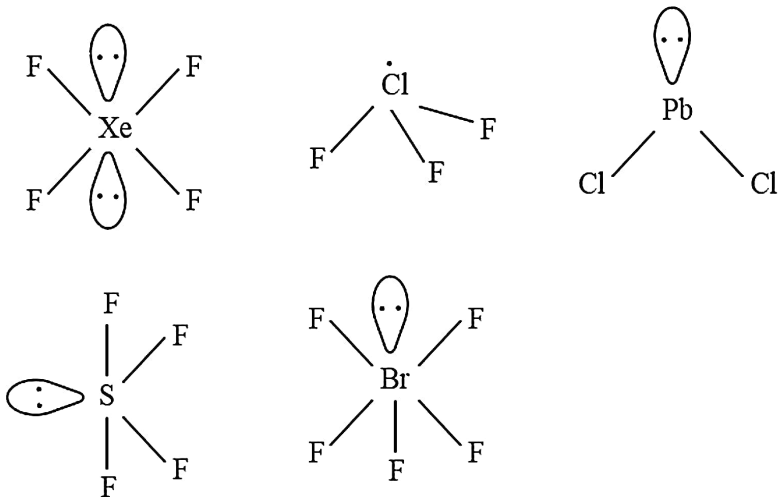
17.76%

SOLUTION

[▶ Video Solution](#)Atomic number (Z) of Cr = 24Electronic configuration = $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^1$ 

Among 6 electrons in $2p$ and $3p$, one electron must have $m = +1$ and $s = -\frac{1}{2}$ but in $3d$ -subshell, an orbital having $m = +1$ may have spin quantum number $-\frac{1}{2}$ or $+\frac{1}{2}$.

Therefore, minimum and maximum possible values of electrons are 2 and 3, respectively.



SOLUTION

[▶ Video Solution](#)

Both (A) and (R) are true but (R) is not the correct explanation of (A)

The melting and boiling points of noble gases are very low in comparison to those of other substances of comparable atomic and molecular masses. This indicates that only weak van der Waals forces or weak London dispersion forces are present between the atoms of the noble gases in the liquid or the solid state.

The van der Waals force increases with the increase in the size of the atom, and therefore, in general, the boiling and melting points increase from He to Rn.

Helium boils at -269°C . Argon has larger mass than helium and have larger dispersion forces. Because of larger size the outer electrons are less tightly held in the larger atoms so that instantaneous dipoles are more easily induced resulting in greater interaction between argon atoms. Therefore, its boiling point (-186°C) is more than that of He.

Similarly, because of increased dispersion forces, the boiling and melting points of monoatomic noble gases increase from helium to radon.

SOLUTION

[▶ Video Solution](#)

For HCl, n -factor is 1

So, for HCl,

Normality = Molarity

$$\text{Molarity} = \frac{\text{Number of moles of HCl}}{\text{Total volume (in L)}}$$

$$\text{Number of moles of HCl} = \text{Molarity} \times \text{Total volume (in L)}$$

$$= 0.1 \times 10 \times 10^{-3} = 10^{-3} \text{ mol}$$

In 1 mol of HCl, the number of HCl molecules present = N_A = Avogadro's number

So, the number of HCl molecules in 10^{-3} moles

$$= 10^{-3} \times N_A = 10^{-3} \times 6.022 \times 10^{23}$$

$$= 6.022 \times 10^{20} \text{ molecules}$$

[< Prev](#)

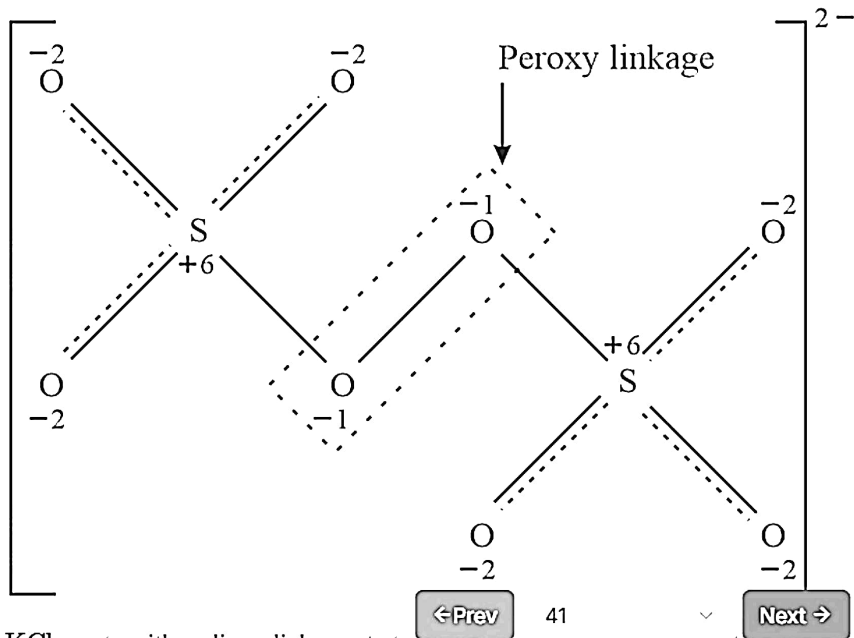
40

v

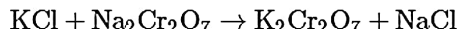
[Next >](#)

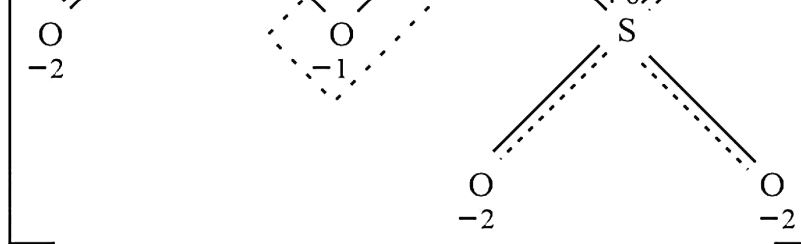
Fe^{2+} reduces $\text{S}_2\text{O}_8^{2-}$ to SO_4^{2-} as it acts as a reducing agent and donates an electron to reach in stable oxidation state +3.

In $\text{S}_2\text{O}_8^{2-}$ there is a peroxy linkage between two oxygen atoms. So those oxygen atoms exhibit -1 oxidation state and all other oxygen atoms exhibit -2 oxidation state. Oxygen atoms which are in -1 oxidation state in $\text{S}_2\text{O}_8^{2-}$ are reduced to -2 oxidation state in SO_4^{2-} .

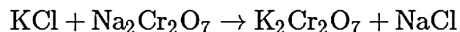


KCl reacts with sodium dichromate to form potassium dichromate. It is not a redox reaction; it is a displacement reaction.





KCl reacts with sodium dichromate to form potassium dichromate. It is not a redox reaction; it is a displacement reaction.



ΔG for $\text{Mn}^{+3}/\text{Mn}^{+2}$ is more negative than for $\text{Cr}^{+3}/\text{Cr}^{+2}$ because Mn^{+3} has the outer electronic configuration of $3d^4$ and Mn^{+2} has the outer electronic configuration of $3d^5$. Thus, the conversion of Mn^{+3} to Mn^{+2} will be a favourable reaction. Since $3d^5$ is a very stable configuration as it is a half-filled configuration. Whereas, Cr^{+3} to Cr^{+2} undergoes a change in outer electronic configuration from $3d^3$ to $3d^4$.

Thus, $-\Delta G_{\text{Mn}^{+3}/\text{Mn}^{+2}} > -\Delta G_{\text{Cr}^{+3}/\text{Cr}^{+2}}$

Vanadium reacts with halogens to form VF_5 , VCl_5 and VBr_3 but does not form VBr_5 or VI_5 because in +5 oxidation state. Vanadium is a strong oxidizing agent, thus converts bromide and iodide to bromine and iodine, respectively. So, VBr_5 and VI_5 are not stable in the air.

EVALUATION

Marked for Review

TIME TAKEN

You: 44 sec

Others: 119 sec

TIME ALLOTTED

90 sec

DIFFICULTY

Medium

CHAPTER

Classification of Elements and Periodicity in Properties

TOPIC

Periodic properties

DIDN'T ATTEMPT ☐

26.62%

GOT IT RIGHT ☒

94.59%

GOT IT WRONG ☐

5.41%

SOLUTION

[▶ Video Solution](#)

For 1 mole of $(\text{Mg} \longrightarrow \text{Mg}^{2+})$

$\text{IE} = \text{IE}_1 + \text{IE}_2 = (740 + 1450) = 2190 \text{ kJ/mol.}$

Number of mole in 4.8 g of $\text{Mg} = 4.8/24 = 0.2 \text{ mol}$

For 1 mole energy required = 2190 kJ/mol

For 0.2 mole energy required = $2190 \times 0.2 = 438 \text{ kJ}$

Thus, for 4.8 g of Mg to Mg^{2+} conversion, energy required is 438 kJ.

Hence, the correct option is (d).

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v

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EVALUATION



TIME TAKEN

You: 186 sec

Others: 60 sec

TIME ALLOTTED

150 sec

DIFFICULTY

Hard

CHAPTER

Haloalkanes and Haloarenes

TOPIC

Preparation and Properties of Aryl Halides

DIDN'T ATTEMPT

17.63%

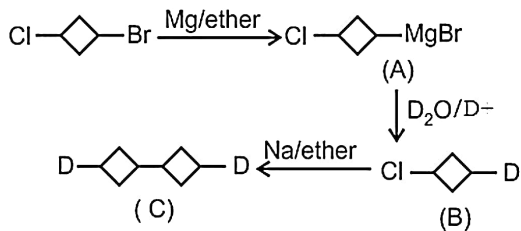
GOT IT RIGHT

83.98%

GOT IT WRONG

16.02%

SOLUTION

[▶ Video Solution](#)[< Prev](#)

43

v

[Next >](#)

wrong, this will help us in providing more accurate analysis.



Misinterpreted Question



Forgot Formula



Overthinking



Missed a Step

EVALUATION

Incorrect

TIME TAKEN

You: 53 sec

Others: 41 sec

TIME ALLOTTED

120 sec

DIFFICULTY

Medium

CHAPTER

Alcohols Phenols and Ethers

TOPIC

Reactions of Alcohols

DIDN'T ATTEMPT ☐

26.50%

GOT IT RIGHT ☒

84.26%

GOT IT WRONG ☐

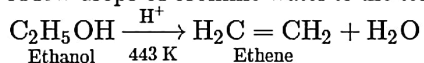
15.74%

SOLUTION

[▶ Video Solution](#)

One of the product in this reaction is ethene, an unsaturated hydrocarbon. It decolourises bromine water. Bromine solution is the classic test for the detection of unsaturation.

A few drops of bromine water to the test tube with the ethanol produce reddish-brown colour.

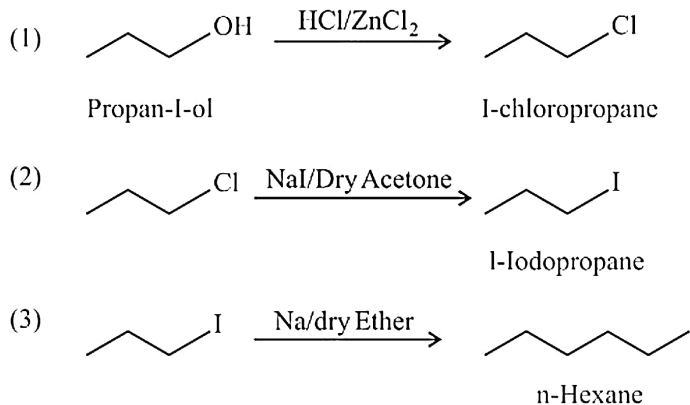


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First step of the reaction involves conversion of propan-1-ol to 1-chloropropane in the presence of lucas reagent(HCl/ZnCl_2)

Second step of the reaction involves halogen exchange by Finkelstein reaction(which involves $\text{S}_{\text{N}}2$ mechanism),Where 1-chloropropane is converted to 1-iodopropane

Third step of the reaction involves wurtz reaction of 1-iodo propane in presence of sodium in dry ether to form n-hexane.

Your Answer

-

Correct Answer

5

EVALUATION

Marked for Review

TIME TAKEN

You: 2 sec

Others: 144 sec

TIME ALLOTTED

180 sec

DIFFICULTY

Hard

CHAPTER

Thermodynamics (C)

TOPIC

Laws of Thermochemistry and Enthalpy Change

DIDN'T ATTEMPT ☐

66.25%

GOT IT RIGHT ☒

80.50%

GOT IT WRONG ☐

19.50%

SOLUTION

▶ Video Solution

Energy released = Energy due to formation of two single bonds = $2 \times 331 = 662 \text{ kJ mol}^{-1}$ of propene

$\Delta H_{\text{polymerisation}} / \text{mol} = 590 - 662 = -72 \text{ kJ mol}^{-1}$

$\Delta H_{\text{polymerisation}} = -72 \times n = -360 \Rightarrow n = 5$

◀ Prev

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▼

Next ▶

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EVALUATION

Marked for review

TIME TAKEN

You: 6 sec

Others: 103 sec

TIME ALLOTTED

120 sec

DIFFICULTY

Easy

CHAPTER

Chemical Kinetics

TOPIC

Rate of Reaction

DIDN'T ATTEMPT ☐

65.05%

GOT IT RIGHT ☒

77.70%

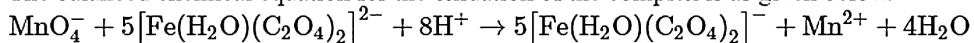
GOT IT WRONG ☐

22.30%

SOLUTION

[▶ Video Solution](#)

The balanced chemical equation for the oxidation of the complex is as given below.



Hence, $-\frac{d}{dt} [\text{MnO}_4^-] = -\frac{1}{8} \frac{d}{dt} [\text{H}^+]$

$$\frac{\frac{d}{dt} [\text{H}^+]}{\frac{d}{dt} [\text{MnO}_4^-]} = 8$$

Hence, the ratio of the rate of change of $[\text{H}^+]$ to the rate of change of $[\text{MnO}_4^-]$ is 8.

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v

[Next >](#)

Given equilibrium reaction: $\text{CaCO}_3 (\text{s}) \rightarrow \text{CaO} (\text{s}) + \text{CO}_2 (\text{g})$

Molar mass of $\text{CaCO}_3 = 100 \text{ g}$

Number of mole of $\text{CaCO}_3 = \frac{10}{100} = 0.1 \text{ mole}$

Concentration of $\text{CaCO}_3 = 0.01 \frac{\text{mole}}{\text{L}}$

Initial concentration: $\underset{0.01}{\text{CaCO}_3} (\text{s}) \rightarrow \underset{0}{\text{CaO}} (\text{s}) + \underset{0}{\text{CO}_2} (\text{g})$

Equilibrium concentration: $\underset{0.01-a}{\text{CaCO}_3} (\text{s}) \rightarrow \underset{a}{\text{CaO}} (\text{s}) + \underset{a}{\text{CO}_2} (\text{g})$

CO_2 does not participate in equilibrium reaction.

The equilibrium constant for given reaction at 1000 K is given as $K = \frac{[\text{CaO}]}{[\text{CaCO}_3]}$.

Given, $K_{1000} = 0.059$

$$0.059 = \frac{a}{0.01-a}$$
$$0.00059 = 1.059 a \Rightarrow a = \frac{0.00059}{1.059}$$
$$a = 0.00056.$$

Hence, concentration of $\text{CaCO}_3 = 0.00939$

Mole of CaCO_3 remain $= 0.00939 \times 10 = 0.0939 \text{ mole}$.

◀ Prev

✓

Next ▶

Initial concentration: $\text{CaCO}_3 \left(\text{s} \right) \rightarrow \text{CaO} \left(\text{s} \right) + \text{CO}_2 \left(\text{g} \right)$

Equilibrium concentration: $\text{CaCO}_3 \left(\text{s} \right) \rightarrow \text{CaO} \left(\text{s} \right) + \text{CO}_2 \left(\text{g} \right)$

CO_2 does not participate in equilibrium reaction.

The equilibrium constant for given reaction at 1000 K is given as $K = \frac{[\text{CaO}]}{[\text{CaCO}_3]}$.

Given, $K_{1000} = 0.059$

$$0.059 = \frac{a}{0.01-a}$$

$$0.00059 = 1.059 a \Rightarrow a = \frac{0.00059}{1.059}$$

$$a = 0.00056.$$

Hence, concentration of $\text{CaCO}_3 = 0.01 - 0.00056 = 0.00939 \frac{\text{mole}}{\text{L}}$.

Mole of CaCO_3 remain $= 0.00939 \times 10 = 0.0939$ mole.

Hence, Mass of CaCO_3 remaining $= 0.0939 \times 100 = 9.39$ g.

SOLUTION

[▶ Video Solution](#)

Given, $\pi = 3.735 \times 10^{-3}$ bar

Mass of $\text{K}_2\text{SO}_4(\omega) = 17.4\text{mg}$

Volume = 2 L

Temperature (T) = $27^\circ\text{C} = 27 + 273 = 300$ K

From osmotic pressure of solution.

$$\pi = iCRT$$

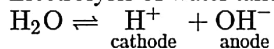
i = van't-Hoff factor

$$i = \pi / CRT$$

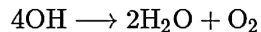
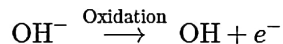
$$= \frac{\pi \times M \times V}{\omega \times R \times T} = \frac{3.735 \times 174 \times 2}{17.4 \times 0.083 \times 300}$$

Molar mass (M) $\text{H}_2\text{SO}_4 = 174, i = 3.0$

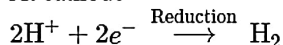
Electrolysis of water takes place as follows



At anode



At cathode



Given, time, $t = 1930 \text{ s}$

Number of moles of hydrogen collected

$$= \frac{1120 \times 10^{-3}}{22.4} \text{ moles}$$

$$= 0.05 \text{ moles}$$

$\therefore$ 1 mole of hydrogen is deposited by = 2 moles of electrons

$\therefore$ 0.05 moles of hydrogen will be deposited by

$$= 2 \times 0.05$$

$$= 0.10 \text{ mole of electrons}$$

Charge,

$$Q = nF$$

$$= 0.1 \times 96500$$

Charge, $Q = it$

$$0.1 \times 96500 = i \times 1930$$

$$i = \frac{0.1 \times 96500}{1930}$$

$$= 5.0 \text{ A}$$

←Prev

50

✓

Next →

DIDN'T ATTEMPT 

59.65%

GOT IT RIGHT 

89.56%

GOT IT WRONG 

10.44%

SOLUTION

[▶ Video Solution](#)

The given equation can be written as

$$(ydx - xdy) + x\sqrt{xy}(x+y)dx + y\sqrt{xy}(x+y)dy = 0$$

$$\Rightarrow (ydx - xdy) + (x+y)\sqrt{xy}(xdx + ydy) = 0$$

$$\Rightarrow \frac{ydx - xdy}{y^2} + \left(\frac{x}{y} + 1\right) \cdot \sqrt{\frac{x}{y}} d\left(\frac{x^2 + y^2}{2}\right) = 0$$

$$\Rightarrow d\left(\frac{x}{y}\right) + d\left(\frac{x^2 + y^2}{2}\right) \left(\frac{x}{y} + 1\right) \sqrt{\frac{x}{y}} = 0$$

$$\Rightarrow d\left(\frac{x^2 + y^2}{2}\right) + \frac{d\left(\frac{x}{y}\right)}{\left(\frac{x}{y} + 1\right) \cdot \sqrt{\frac{x}{y}}} = 0$$

Integrating both the sides, we get

$$\Rightarrow \frac{x^2 + y^2}{2} + 2\tan^{-1} \sqrt{\frac{x}{y}} = C$$

EVALUATION

Marked for review

TIME TAKEN

You: 4 sec

Others: 162 sec

TIME ALLOTTED

150 sec

DIFFICULTY

Medium

CHAPTER

Vector Algebra

TOPIC

Product of 2 vectors

DIDN'T ATTEMPT ☐

15.35%

GOT IT RIGHT ☒

86.62%

GOT IT WRONG ☐

13.38%

SOLUTION

[▶ Video Solution](#)

$$\vec{r} \cdot \hat{i} = 2\vec{r} \cdot \hat{j} = 4\vec{r} \cdot \hat{k}$$

$$\text{Let } \vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

$$x = 2y = 4z$$

$$|\vec{r}| = \sqrt{84} \Rightarrow \sqrt{(4z)^2 + (2z)^2 + z^2} = \sqrt{84}$$

$$z = 2; y = 4; x = 8$$

$$\vec{r} = 8\hat{i} + 4\hat{j} + 2\hat{k}$$

$$\vec{r} \cdot (2\hat{i} - 3\hat{j} + \hat{k}) = 16 - 12 + 2 = 6$$

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harsh

The equation of line is,

$$\frac{x-4}{3} = \frac{y-2}{4} = \frac{z-4}{-5} = t \quad \dots [Given]$$

A point of line is $(3t + 4, 4t + 2, -5t + 4)$

$$= B, (\text{say})$$

Direction ratios of $\overrightarrow{AB}$ are, $(3t + 2), (4t - 1), (-5t + 5)$

Since, $\overrightarrow{AB}$ is perpendicular to the line

$$\frac{x-3}{3} = \frac{y-8}{1} = \frac{z-3}{1}$$

$$\Rightarrow 3(3t + 2) + 1(4t - 1) + 1(-5t + 5) = 0$$

$$\Rightarrow 8t + 10 = 0$$

$$\Rightarrow 4t = -5$$

$$\Rightarrow |4t| = 5$$

$$\begin{array}{ccc} 2 & & 3 \\ \hline A(0,4) & | & P & B(2\cos \theta, 2\sin \theta) \end{array}$$

By using section formula:

$$\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$$

$$P\left(\frac{4\cos\theta+0}{5}, \frac{4\sin\theta+12}{5}\right) \rightarrow (h, k)$$

$$\Rightarrow \cos\theta = \frac{5h}{4}, \sin\theta = \frac{5k-12}{4}$$

$$\Rightarrow \cos^2\theta + \sin^2\theta = 1$$

$$\Rightarrow \left(\frac{5h}{4}\right)^2 + \left(\frac{5k-12}{4}\right)^2 = 1$$

$$\Rightarrow (5x)^2 + (5y - 12)^2 = 16$$

(This is the equation of circle)

EVALUATION

Incorrect

TIME TAKEN

You: 528 sec

Others: 211 sec

TIME ALLOTTED

210 sec

DIFFICULTY

Hard

CHAPTER

Binomial Theorem

TOPIC

Terms of Binomial Expansion

DIDN'T ATTEMPT ○

38.01%

GOT IT RIGHT ✓

81.53%

GOT IT WRONG ✗

18.47%

SOLUTION

[▶ Video Solution](#)

$$\begin{aligned}
 & (1 - x^3)^5 (1 + x^2)^4 (1 + x^4)^8 \\
 & \left[{}^5C_0 - {}^5C_1x^3 + {}^5C_2x^6 - \dots \right] \left[{}^4C_0 + {}^4C_1x^2 + {}^4C_2x^4 + {}^4C_3x^6 + {}^4C_4x^8 \right] \\
 & \left[{}^8C_0 + {}^8C_1x^4 + \dots \right] \\
 & \text{Coefficient of } x^6 \\
 & = {}^5C_0 {}^4C_3 {}^8C_0 + {}^5C_0 {}^4C_1 {}^8C_1 + {}^5C_2 {}^4C_0 {}^8C_0 \\
 & = 4 + 32 + 10 = 46
 \end{aligned}$$

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Activate Windows
Go to Settings to activate Windows.

DIFFICULTY
MediumCHAPTER
Sequences and SeriesTOPIC
Geometric ProgressionDIDN'T ATTEMPT ☐
34.59%GOT IT RIGHT ☒
88.30%GOT IT WRONG ☐
11.70%

SOLUTION

[▶ Video Solution](#)For the G.P. $a, ar, ar^2, \dots$

$$P_n = a(ar)(ar^2) \dots (ar^{n-1})$$

$$= a^n r^{\frac{n(n-1)}{2}}$$

$$\Rightarrow \sqrt[n]{P_n} = ar^{\left(\frac{n-1}{2}\right)}$$

$$\Rightarrow \sum_{n=1}^{\infty} \sqrt[n]{P_n} = a[1 + \sqrt{r} + r + r\sqrt{r} + \dots]$$

$$= \frac{a}{1-\sqrt{r}}$$

Given, $a = 16$ and $r = \frac{1}{4}$

$$= \frac{16}{1-\sqrt{\frac{1}{4}}} = 32$$

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EVALUATION

Not Answered

TIME TAKEN

You: 5 sec

Others: 208 sec

TIME ALLOTTED

150 sec

DIFFICULTY

Easy

CHAPTER

Differential Equations

TOPIC

Homogeneous DE

DIDN'T ATTEMPT ☐

32.49%

GOT IT RIGHT ☒

91.47%

GOT IT WRONG ☐

8.53%

SOLUTION

[▶ Video Solution](#)

$$\begin{aligned}\frac{dy}{dx} &= \frac{y + \sqrt{x^2 + y^2}}{x}, y = vx \Rightarrow \frac{xdv}{dx} + v = v + \sqrt{1 + v^2} \\ \Rightarrow \int \frac{dv}{\sqrt{1 + v^2}} &= \int \frac{dx}{x} \Rightarrow \ln(v + \sqrt{1 + v^2}) = \ln x + \ln c \\ \Rightarrow v + \sqrt{1 + v^2} &= cx \Rightarrow y + \sqrt{x^2 + y^2} = cx^2\end{aligned}$$

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DIDN'T ATTEMPT 

51.44%

GOT IT RIGHT 

67.67%

GOT IT WRONG 

32.33%

SOLUTION

[▶ Video Solution](#)

Let $x^2 - 5x + 4 = a \quad \dots (1)$

$$\therefore a(y^2 + y + 1) < 2y$$

$$\Rightarrow ay^2 + y(a - 2) + a < 0 \text{ for all } y \in R$$

$$\therefore a < 0 \text{ and } D < 0$$

$$\Rightarrow a < 0 \text{ and } (a - 2)^2 - 4a^2 < 0$$

$$\Rightarrow 3a^2 + 4a - 4 > 0$$

$$\therefore a < 0 \text{ and } a < -2$$

$$\Rightarrow a < -2 \quad \dots (ii)$$

From Eqs. (i) and (ii), $x^2 - 5x + 4 < -2$

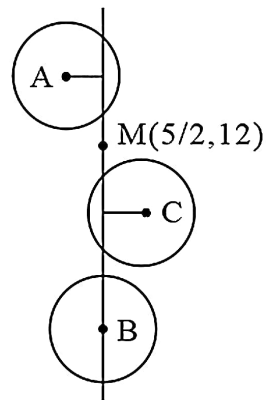
$$\Rightarrow x^2 - 5x + 6 < 0 \Rightarrow x \in (2, 3)$$

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$A(0, 16), B(3, 0), C(5, 8)$

line will pass through the mid point of A & C

$$m_{MB} = \frac{12}{-1/2} = -24$$

$$\left| \frac{m}{8} \right| = 3.$$

EVALUATION

Marked for review

TIME TAKEN

You: 21 sec

Others: 95 sec

TIME ALLOTTED

180 sec

DIFFICULTY

Hard

CHAPTER

Probability

TOPIC

Classical Definition of Probability

DIDN'T ATTEMPT ☐

65.17%

GOT IT RIGHT ☒

78.86%

GOT IT WRONG ☐

21.14%

SOLUTION

[▶ Video Solution](#)

The last digit of the product will be 1, 2, 3, 4, 6, 7, 8 or 9 if and only if each of the n positive integers ends in any of these digits. Now the probability of an integer ending

in 1, 2, 3, 4, 6, 7, 8 or 9 is $\frac{8}{10}$. Therefore the probability that the last digit of the

product of n integer in 1, 2, 3, 4, 6, 7, 8 or 9 is $\left(\frac{4}{5}\right)^n$. The probability for an integer to

end in 1, 3, 7 or 9 is $\frac{4}{10} = \frac{2}{5}$

Therefore the probability for the product of n positive integers to end in 1, 3, 7 or 9 is $\left(\frac{2}{5}\right)^n$

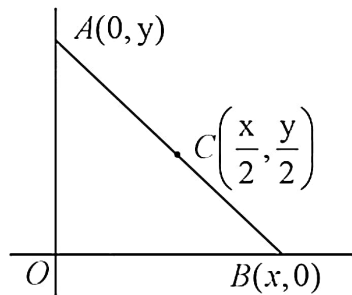
Hence the required probability = $\left(\frac{4}{5}\right)^n - \left(\frac{2}{5}\right)^n = \frac{4^n - 2^n}{5^n}$

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Let the stick be $AB = a$ cm.

Coordinates of mid-point of AB is $D\left(\frac{x}{2}, \frac{y}{2}\right)$.

Velocity of point B is $b \frac{\text{cm}}{\text{s}}$. Therefore,

$$\frac{dx}{dt} = b$$

Now, in $\triangle AOB$

$$AB^2 = OA^2 + OB^2$$

$$\Rightarrow a^2 = x^2 + y^2$$

Differentiating both sides, we have

$$2a \frac{da}{dt} = 2x \frac{dx}{dt} + 2y \frac{dy}{dt}$$

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✓

Next →

$$2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0$$

$$\Rightarrow x \frac{dx}{dt} + y \frac{dy}{dt} = 0$$

$$\Rightarrow bx + y \frac{dy}{dt} = 0$$

When the stick is equally inclined then $x = y$

$$by + y \frac{dy}{dt} = 0$$

$$\Rightarrow \frac{dy}{dt} = -b$$

Now coordinates of mid-point of AB is $D\left(\frac{x}{2}, \frac{y}{2}\right)$. Therefore, velocity vector at D is

$$\frac{1}{2} \left(\frac{dx}{dt} \right) + \frac{1}{2} \left(\frac{dy}{dt} \right).$$

Resultant velocity of mid point of AB is

$$\sqrt{\left(\frac{1}{2} \cdot \frac{dx}{dt}\right)^2 + \left(\frac{1}{2} \cdot \frac{dy}{dt}\right)^2}$$

$$= \sqrt{\frac{b^2}{4} + \frac{b^2}{4}}$$

$$= \frac{b}{\sqrt{2}} \frac{\text{cm}}{\text{s}}$$

◀ Prev

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∨

Next ▶



16.63% marked this



Correct

You Marked

72.75% marked this

EVALUATION



TIME TAKEN

You: 92 sec

Others: 165 sec

TIME ALLOTTED

120 sec

DIFFICULTY

Easy

CHAPTER

Trigonometric Ratios & Identities

TOPIC

Trigonometric Series

DIDN'T ATTEMPT ☐

57.01%

GOT IT RIGHT ☒

72.75%

GOT IT WRONG ☐

27.25%

SOLUTION

[▶ Video Solution](#)

$$\text{Let } \theta = \left(\frac{90}{17}\right)^\circ \Rightarrow 17\theta = 90^\circ$$

$$\begin{aligned} \text{Now, } 32 \sin \frac{90^\circ}{17} \cos \frac{180^\circ}{17} \cos \frac{360^\circ}{17} \cos \frac{720^\circ}{17} &= 32 \sin \theta \cos 2\theta \cos 4\theta \cos 8\theta = \frac{32 \cos \theta \sin \theta \cos 2\theta \cos 4\theta \cos 8\theta}{\cos \theta} = 2 \times \frac{\sin 16\theta}{\cos \theta} \\ &= \frac{2 \sin(17\theta - \theta)}{\cos \theta} = \frac{2 \sin(90^\circ - \theta)}{\cos \theta} = 2 \end{aligned}$$

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SOLUTION

[▶ Video Solution](#)

lies $\alpha = a + ib, \gamma = c + id$

$$\sqrt{a^2 + b^2} + \sqrt{c^2 + d^2} = p$$

$$f(z) = (4 + i)z^2 + z(a + ib) + (c + id)$$

$$f(1) = (4 + a + c) + i(1 + b + d)$$

hence $1 + b + d = 0 \dots (1)$

$$f(i) = -(4 + i) + i(a + ib) + (c + id)$$

$$-4 - b + c + i(a + d - 1)$$

hence $a + d = 1 \dots (2)$

from (1) & (2) $a - b = 2$

hence there is no restriction of 'c' let $c = 0$

$$|\alpha| + |\gamma| = \sqrt{a^2 + b^2} + \sqrt{d^2} = \sqrt{4 + 2ab} + |d| \leq \sqrt{4 + 2ab} \geq \sqrt{2}$$

with equality

$$d = 0, a = 1, b = -1$$

$$\text{hence } p = \sqrt{2}$$

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EVALUATION

Marked for Review

TIME TAKEN

You: 50 sec

Others: 97 sec

TIME ALLOTTED

240 sec

DIFFICULTY

Hard

CHAPTER

Functions

TOPIC

Types of Function (Mapping)

DIDN'T ATTEMPT ☐

64.09%

GOT IT RIGHT ☒

54.89%

GOT IT WRONG ☐

45.11%

SOLUTION

[▶ Video Solution](#)

As $f(x)$ is one-one function, so $f(x^3 + 14x^2 + 13x - 5) = f(1 - x^2 + x^3)$

$$\Rightarrow x^3 + 14x^2 + 13x - 5 = 1x^3 + x^3 \Rightarrow 15x^2 + 13x - 6 = 0$$

$$\Rightarrow 15x^2 + 18x - 5x - 6 = 0 \Rightarrow (3x - 1)(5x + 6) = 0$$

$$\Rightarrow x = \frac{1}{3}, \frac{-6}{5} \text{ but } x \in [-1, 1]$$

Hence $x = \frac{1}{3}$ only.

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v

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EVALUATION

Not Answered

TIME TAKEN

You: 13 sec

Others: 122 sec

TIME ALLOTTED

180 sec

DIFFICULTY

Medium

CHAPTER

Three Dimensional Geometry

TOPIC

Line in Space

DIDN'T ATTEMPT ☐

14.15%

GOT IT RIGHT ☒

85.37%

GOT IT WRONG ☐

14.63%

SOLUTION

[▶ Video Solution](#)

Any point on the first line is $(4r + k, 2r + 1, r - 1)$, and any point on the second line is $(r' + k + 1, -r', 2r' + 1)$ for some values of r and r' . The lines are intersecting if these two points coincide i.e

$$4r + k = r' + k + 1, 2r + 1 = -r', r - 1 = 2r' + 1 \text{ for some } r \text{ and } r'$$

$$\Rightarrow 4r - r' = 1, 2r + r' = -1, r - 2r' = 2$$

$$\text{Now, } 4r - r' = 1, 2r + r' = -1 \Rightarrow r = 0, r' = -1 \text{ which satisfy } r - 2r' = 2.$$

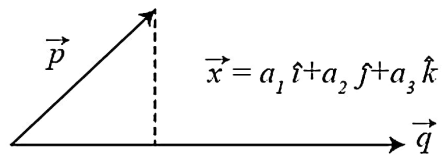
$\Rightarrow$ The given lines are intersecting for all real values of k .

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Component of $\vec{p}$ in the direction of $\vec{q}$

$$= \frac{(\vec{p} \cdot \vec{q})\vec{q}}{|\vec{q}|^2} = \frac{-5\hat{i} + 5\hat{j} - 5\hat{k}}{3}$$

Component of $\vec{p}$, perpendicular to

$$\begin{aligned} \vec{q} &= \vec{p} - \left(\frac{-5\hat{i} + 5\hat{j} - 5\hat{k}}{3} \right) \\ &= (2\hat{i} + 4\hat{j} + 3\hat{k}) - \left(\frac{-5\hat{i} + 5\hat{j} - 5\hat{k}}{3} \right) \\ &= \frac{11}{3}\hat{i} + \frac{7}{3}\hat{j} - \frac{4}{3}\hat{k} \end{aligned}$$

Please select the reason why you attempted the question wrong, this will help us in providing more accurate analysis.

-  Silly Mistake  Calculation Error  Time Pressure  Conceptual Gap
-  Misinterpreted Question  Forgot Formula  Overthinking  Missed a Step

EVALUATION

Incorrect

TIME TAKEN

You: 59 sec
Others: 114 sec

TIME ALLOTTED

120 sec

DIFFICULTY

Easy

CHAPTER

Matrices

TOPIC

Adjoint and its Properties

DIDN'T ATTEMPT 

43.94%

GOT IT RIGHT 

49.65%

GOT IT WRONG 

50.35%

SOLUTION

 Video Solution

$$\begin{aligned}A^{-1} + B^{-1} &= 2I \\ \Rightarrow I + AB^{-1} &= 2A \\ \Rightarrow B + A &= 2AB \text{ or } 2AB = I\end{aligned}$$

← Prev

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✓

Next →

Activate Windows
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we have focus $(-1, -1)$ and corresponding directrix $x - y + 3 = 0$ $e = \frac{1}{2}$ (given)

Now, equation of axis of ellipse is

$$x + y + k = 0$$

As (1) passes through $(-1, -1)$, so $k = 2$

∴ Equation of axis is

$$x + y + 2 = 0 \dots\dots (3)$$

∴ on solving (1) and (3), we get foot of directrix

$$\left(\frac{-5}{2}, \frac{1}{2}\right)$$

Now, distance between focus and corresponding foot of directrix

$$= \left(\frac{a}{e} - ae\right) = \sqrt{\left(\frac{-5}{2} + 1\right)^2 + \left(\frac{1}{2} + 1\right)^2} \quad \left(\text{as } e = \frac{1}{2}\right)$$

$$\Rightarrow 2a - \frac{a}{2} = \frac{3}{\sqrt{2}} \Rightarrow a = \sqrt{2}$$

Also equation of mirror axis is

$$x - y + \lambda = 0$$

As distance of minor axis from focus is ae , so

$$\frac{|-1+1+\lambda|}{\sqrt{2}} = \sqrt{2}\left(\frac{1}{2}\right) \Rightarrow |\lambda| = 1 \Rightarrow \lambda = \pm 1$$

To find the number of solutions of the following system of linear homogeneous equations.

$$x - y + z = 0 \dots (i)$$

$$x + 2y - z = 0 \dots (ii)$$

$$2x + y + 3z = 0 \dots (iii)$$

We can write the above equations in matrix form i.e.

$$\begin{bmatrix} 1 & -1 & 1 \\ 1 & 2 & -1 \\ 2 & 1 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$$\text{or } AX = 0, \text{ where } A = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 2 & -1 \\ 2 & 1 & 3 \end{bmatrix}$$

Since, we know that, if $|A| = 0$, then only a non-trivial solution exist.

$$\text{Consider, } |A| = \begin{vmatrix} 1 & -1 & 1 \\ 1 & 2 & -1 \\ 2 & 1 & 3 \end{vmatrix}$$

$$\therefore |A| = 1[2 \times 3 - (-1)] + 1[3 + 2] + 1[1 - 4]$$

$$= 7 + 5 - 3 = 9 \neq 0$$

$$\therefore |A| \neq 0$$

Hence, there is no non-trivial solutions. $\Rightarrow$ There will be only 1 trivial solution.

We have, $f\left(\frac{2x+3}{3x+5}\right) = x + 4$

$$\text{Put, } \frac{2x+3}{3x+5} = y \Rightarrow 2x+3 = 3xy+5y \Rightarrow x = \frac{5y-3}{2-3y}$$

$$\therefore f(y) = \frac{5y-3}{2-3y} + 4 = \frac{5y-3+8-12y}{2-3y} \Rightarrow f(y) = \frac{7y-5}{3x-2}$$

$$\therefore f(x) = \frac{7x-5}{3x-2} \Rightarrow \int f(x)dx = \int \frac{7x-5}{3x-2} dx$$

$$\text{Put, } 3x-2 = t \Rightarrow x = \frac{t+2}{3}$$

$$dx = \frac{dt}{3} = \frac{1}{3} \int \frac{7\left(\frac{t+2}{3}\right) - 5}{t} dt$$

$$= \frac{1}{9} \int \frac{7t-1}{t} dt = \frac{7}{9} \int dt - \frac{1}{3} \int \frac{dt}{t}$$

$$= \frac{7}{9} t - \frac{1}{9} \log t + C$$

$$= \frac{7}{9} (3x-2) - \frac{1}{9} \log |3x-2| + C$$

$$= \frac{7}{3} x - \frac{1}{9} \log |3x-2| + C$$

$$\text{Here, } A = \frac{7}{3}, B = -\frac{1}{9}$$

$$\therefore 3B - A = 3\left(-\frac{1}{9}\right) - \frac{7}{3} = -\frac{8}{3}$$

◀ Prev

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EVALUATION

Marked for Review

TIME TAKEN

You: 4 sec

Others: 111 sec

TIME ALLOTTED

210 sec

DIFFICULTY

Hard

CHAPTER

Permutation Combination

TOPIC

Selection of one or more item

DIDN'T ATTEMPT ☐

54.26%

GOT IT RIGHT ☒

58.33%

GOT IT WRONG ☐

41.67%

SOLUTION

[▶ Video Solution](#)

In the string there are five , four , three , two and one

Case(I): Unique letters

Case(ii): Same and Different

Case(iii): Same, More Same and Unique

Case(iv): same, Unique

case (v): same, same

Case(vi): same, unique

case(vii): all same

Hence, total number of ways

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$f(x) = \frac{1}{2} - x + \frac{1}{2} [2x] - \frac{1}{2} [1 - 2x]$ for $x \in [0, 1]$, $2x \in [0, 2]$ and $2x$ takes integer values 0, 1, 2 at $x = 0, \frac{1}{2}, 1$. Also for $x \in [0, 1]$; $(1 - 2x) \in [-1, 1]$ and $(1 - 2x)$ takes integer values -1, 0, 1 at $x = 1, \frac{1}{2}$ and 0 respectively. So, we shall break the interval $[0, 1]$ at $x = 1$

$$\therefore f(x) = \begin{cases} \frac{1}{2} - 0 + 0 - \frac{1}{2}; & x = 0 \\ \frac{1}{2} - x + \frac{1}{2} (0) - \frac{1}{2} (0); & 0 < x < \frac{1}{2} \\ \frac{1}{2} - \frac{1}{2} + \frac{1}{2} (1) - \frac{1}{2} (0); & x = \frac{1}{2} \\ \frac{1}{2} - x + \frac{1}{2} (1) - \frac{1}{2} (-1); & \frac{1}{2} < x < 1 \\ \frac{1}{2} - 1 + \frac{1}{2} (2) - \frac{1}{2} (-1); & x = 1 \end{cases}$$

$$\Rightarrow f(x) = \begin{cases} 0; & x = 0 \\ \frac{1}{2} - x; & 0 < x < \frac{1}{2} \\ \frac{1}{2}; & x = \frac{1}{2} \\ \frac{3}{2} - x; & \frac{1}{2} < x < 1 \\ 1; & x = 1 \end{cases}$$

$\therefore f(x)$ is discontinuous at $x = 0, \frac{1}{2}$ and 1 and

$$f'(x) = \begin{cases} -1; & 0 < x < \frac{1}{2} \\ -1; & \frac{1}{2} < x < 1 \end{cases}$$

$\Rightarrow f(x)$ is non-differentiable and discontinuous at $x = 0, \frac{1}{2}$ and 1 i.e., at 3 points.

EVALUATION

Not Answered

TIME TAKEN

You: 13 sec

Others: 163 sec

TIME ALLOTTED

180 sec

DIFFICULTY

Hard

CHAPTER

Binomial Theorem

TOPIC

Multinomial Theorem

DIDN'T ATTEMPT ☐

51.74%

GOT IT RIGHT ☒

92.53%

GOT IT WRONG ☐

7.47%

SOLUTION

[▶ Video Solution](#)

Consider the given expression

$$(abc + abd + acd + bcd)^{10} = a^{10}b^{10}c^{10}d^{10} \left[\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d} \right]^{10}$$

Coefficient of $a^8b^4c^9d^9$ in $(abc + abd + acd + bcd)^{10}$

$$= \text{Coefficient of } a^{-2}b^{-6}c^{-1}d^{-1} \text{ in } \left[\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d} \right]^{10}$$

$$= \frac{10!}{2!6!1!1!}$$

$$= \frac{10 \times 9 \times 8 \times 7}{2}$$

$$= 2520$$

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v

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SOLUTION

[▶ Video Solution](#)

$$f''(x) = \lambda(x-1)$$

Integrating both the side with respect to x

$$\Rightarrow f'(x) = \lambda \frac{(x-1)^2}{2} + c$$

$$\text{As } f'(1) = 0 \Rightarrow c = -2\lambda$$

$$f'(x) = \lambda \frac{(x-1)^2}{2} - 2\lambda$$

Integrating both the side with respect to x

$$f(x) = \frac{\lambda(x-1)^3}{6} - 2\lambda x + u$$

$$f(1) = -6 \Rightarrow -2\lambda + u = -6$$

$$f(-1) = 10 \Rightarrow \frac{2\lambda}{3} + u = 10$$

$$\therefore \lambda = 6 \text{ and } u = 6$$

$$f'(x) = 0 \Rightarrow x - 1 \text{ or } x = 3$$

$$f''(3) > 0$$

$\therefore x = 3$, is a point of local minima.

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The number of real values of m for

$$A = \{x : (x - 2)(x + m + 1) = 0, x \in R\}$$

$$B = \{x : (m - 1)x^2 + mx + 1 = 0, x \in R\}$$

Case I.

$$\text{When } m - 1 = 0 \quad m = 1$$

$$A = \{2, -2\} \text{ and } B = \{-1\}$$

$$\therefore A \cup B = \{2, -2, -1\} \implies \text{exactly 3 elements}$$

$$\therefore m = 1$$

Case - II

$$\text{When } m \neq 1$$

$$A = \{2, -m - 1\}, B = \left\{-1, \frac{-1}{m-1}\right\}$$

$$2 = -m - 1 \Rightarrow m = -3$$

$$A = \{2\}, B = \left\{-1, \frac{1}{4}\right\}$$

$$\therefore m = -3$$

$$2 = \frac{-1}{m-1} \Rightarrow m = \frac{1}{2}$$

$$A = \left\{2, \frac{3}{2}\right\}, B = \{-1, 2\}$$

$$\therefore m = \frac{1}{2}$$

$$-m - 1 = \frac{-1}{m-1} \Rightarrow m^2 - 1 = 1$$

$$\therefore m = \pm\sqrt{2}$$

$$-m - 1 = -1 \Rightarrow m = 0$$

$$-1 = \frac{-1}{m-1} \Rightarrow m - 1 = 1$$

$$\therefore m = 2$$

$$\text{Number of values of } m = 7$$

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✓

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